

How Ride-Sharing Is Shaping Public Transit System: A Counterfactual Estimator Approach

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The new sharing economy model has introduced a dramatic, disruptive impact on the traditional industries by matching the demand and supply in real time. In this study, we examine how the entry of Uber, a ride-sharing services digital platform, brings new disruptive changes in public transportation operations. Significant debate has surrounded whether the new ride-sharing model siphoned riders from public transit or made public transit feasible for more riders, but no consensus has been reached. One reason could be that the commonly used difference-in-differences empirical strategy fails to account for time-varying unobserved confounders. To address this issue, we introduce a class of counterfactual estimators (CEs) to strengthen our causal identification and perform diagnostic tests to validate model assumptions for each CE. A significant drop in passenger trips with buses is found after Uber entry from both the conventional two-way fixed effects model and the CEs. Moreover, we provide empirical evidence that the entry of Uber has not significantly affected public demand response transportation, which indicates that Uber is not directly competing with other transportation services that aim to solve the “last-mile problem.” Last, our additional analyses suggest that the effect of Uber entry is not uniform for different urban areas. All these empirical findings can be coherently explained in our framework of the substitution effect of Uber.

Key words: disruptive technology; public transportation operations; sharing economy; ride-sharing platform; counterfactual estimators

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1. Introduction

1.1. Motivations

Sharing economy is an economic model often defined as a peer-to-peer based activity of acquiring, providing, or sharing access to goods and services that are facilitated by a community-based online platform (Benjaafar and Hu 2020, Benjaafar et al. 2019, He et al. 2021, Hu 2021, Wang et al. 2021b). The idea that people have shared the use of assets is not new, but the advent of the Internet—together with its use of big data—has made it easier for asset owners and those seeking to use the assets to find each other through online digital sharing economy platforms. This sort of dynamics is also referred to as the collaborative consumption, collaborative economy (Belk 2014), peer economy (Bellotti et al. 2015), or gig-economy platforms, which enable a flexible work arrangement (Burtch et al. 2018). Sharing economies allow individuals and groups to make money from underused assets. In this way, physical assets are shared as services. The digital technology underlying the concept of sharing economy has brought new disruptive

changes in operations management (OM) of traditional industries (Olsen and Tomlin 2020, Tian and Jiang 2018).

In recent years, the platform-based sharing economy has received tremendous attention from both industrial and academic research. There is a substantial potential supply for ride-sharing services such as Lyft and Uber. According to a report by Brookings Institute (Yaraghi and Ravi 2017), private vehicles remain unused for 95% of their lifetime. The same report also analyzes the cost advantage of Airbnb over the hotel space as homeowners' leverage spare bedrooms. Airbnb rates were reported to be between 30% and 60% cheaper than hotel rates around the world. In academic research, numerous studies have investigated the question of how platform-based sharing economies pose disruptive challenges for traditional industry operations. Zervas et al. (2017) estimated the impact of Airbnb on the hotel industry and found that the impact is large but uneven across the industry, with lower end hotels being the most affected. Cramer and Krueger (2016) documented several advantages of the Uber platform over traditional taxi services.

One early pioneer and also a significant part of the sharing economy is the online ride-sharing platforms, such as Uber, founded in 2009. There is great interest in the changes that the sharing economy platform brings to various outcomes, such as road driving safety (Greenwood and Wattal 2017), traffic congestion (Li et al. 2021b), local durable good consumption (Gong et al. 2017), and local entrepreneurial activity (Burtch et al. 2018). However, one of the crucial aspects of a ride-sharing platform is how it changes public transportation operations: How the ride-sharing platform influences or coexists with traditional transportation systems. Cramer and Krueger (2016) compared the Uber platform vs. traditional Taxi services, and concluded that UberX drivers achieve a significantly higher capacity utilization rate than taxi drivers. To date, evidence regarding how ride-sharing platforms impact other transportation operations, especially the public transportation system, is incongruous.

1.2. Research Questions and Potential Challenge

Has Uber siphoned riders from public transit, or has it made public transit feasible for more riders? The answers are still open to debate and inconclusive. On the one hand, ride-sharing services, such as Uber and Lyft, can complement public transit by connecting riders to transportation hubs and help finish the last-mile transportation. Ride-sharing services help people travel the “last mile” between transit hubs (e.g., bus stops) and their homes. For example, Uber claims the partnerships with regional transportation authorities are part of its mission “to increase access to transportation.” The partnerships between traditional public transportation agencies and Uber and Lyft have boomed since 2016. In February 2016, the “Direct Connect” program was launched in St. Petersburg, Florida: The transportation agencies gave \$5 discounts on rides provided by Uber and Lyft to and from 24 popular bus stops in its service area (Citylab 2018). Uber has partnered with Masabi, a company in mobile ticketing for public transit, to allow riders to buy public transit fares in its app (Masabi 2018). In addition to solving the “last-mile challenge,” ride-sharing services reduce the need for car ownership. Uber has called itself and public transit a winning combination, which makes it possible to complete desired trips without owning a car (UberBlog 2017). Whether Uber’s effort will do much to boost the public transit system is not clear, but it seems less likely to hurt.

However, on the other hand, ride-sharing services can also compete with public transit systems by providing an alternative mode of transportation. In particular, ride-sharing services are more convenient and cheaper than taxis, and hence are more likely to be a

substitutive product to public transit. In economic theory, substitutes are products that a consumer perceives as similar or comparable and may replace each other, and there is a negative relationship in demand changes among substitutive products (Kwark et al. 2021, Mas-Colell et al. 1995). In practice, Clewlow and Mishra (2017) found that ride-sharing services attract passengers away from public transit, biking, and walking. Experts suspect that the national trend of falling bus ridership could be partially due to the growing popularity of ride-sharing services (Citylab 2019, Streetsblog USA 2019, The Washington Post 2016). Given the “massive market opportunity” in replacing bus and subway rides, there is no guarantee that Uber and Lyft will not draw a privileged subset of passengers away from public transit systems. That, in turn, undermines support for public transportation (The New York Times 2019). Recently, transportation and urban planning experts were worried when Uber released its S-1 regulatory filing in 2019, saying its growth depends on better competing with public transportation (CNN Business 2019).

Therefore, both directions are theoretically plausible and practically ambiguous, which provides a viable opportunity for empirical research. In order to examine the causal influence of Uber’s entry, following the literature examining the ride-sharing platform’s impact, we start with leveraging the introduction of the ride-sharing services by Uber into urban areas (UAs) in the United States between 2011 and 2015 as a natural experiment that exogenously shifts demand for ride-sharing services for local UAs. These changes provide an empirical identification strategy that allows us to execute the commonly used empirical difference-in-differences (DID) estimation to compare the demand for the public transportation system before and after Uber’s entry. The main challenge of applying DID estimation is to ensure the assumption of no unobserved time-varying confounders (also called “parallel trends”), which states that in the absence of the treatment, the average outcomes of treated and control units would have followed parallel paths (Xu 2017). However, this assumption is not directly testable. Instead, a common practice is to ensure pretreatment parallel trends by constructing treated and control units conditioning on pretreatment observables using matching methods. Researchers have more confidence in the validity of the “parallel trends” assumption when they find that the average outcomes of the treated and control units follow parallel paths in pretreatment periods. However, this could not rule out that the presence of unobserved time-varying confounders causes the failure of this assumption.

To overcome this challenge, we introduce a set of recently developed counterfactual estimators (CEs),

which are rooted in the econometrics literature on factor models and interactive fixed effects (IFect; Liu et al. 2020), and computer science and statistics literature on matrix completion (MC; Athey et al. 2021). They include (i) the fixed-effect CE (FEct), (ii) the interactive fixed-effect CE, and (iii) the MC estimator. They directly impute the missing potential observations—posttreatment outcomes of treated units in the absence of the treatment in time-series cross-sectional setting with dichotomous treatments. However, they differ in the underlying outcome model for predicting treated counterfactuals. They provide more reliable causal estimates than conventional two-way fixed effects models when the treatment effect is heterogeneous or there exist unobserved time-varying confounders (Liu et al. 2020). Intuitively, we use the number of public bus trips without Uber's influence to predict the counterfactual number of public bus trips for treated UAs in the periods after Uber entry observed. Then, we obtain average treatment effect on treated (ATT) by taking an average of the differences between observed outcomes and counterfactual (predicted) outcomes for the treated observations.

The main sample is assembled in the following fashion. We first adopt data on Uber's entries on all major UAs from Li et al. (2021b), which conduct research with the same UA level analyses. For each UA, Li et al. (2021b) search the official Uber newsroom as well as the major news media to find out if and when Uber entered a UA. Next, we obtain the rest of our data from multiple resources. To measure the demand for public transportation, we retrieved the unlinked bus passenger trips data in the National Transit Database (NTD) from Federal Transit Administration. We downloaded the monthly ridership time-series report, which contains monthly-updated service information reported by more than 300 UAs from 2006 to 2017. Further, we control a rich set of variables on fuel cost from the U.S. Energy Information Administration and the economic status characteristics of UAs from the Bureau of Economic Analysis. Our moderating variables are collected from the United States Census Bureau, and we primarily focus on the demographic and socioeconomic characteristics of UAs when Uber entered. Moderating factors are examined from four aspects: commuting mode to work, disability and language barrier, unemployment, and poverty rate. After assembling different sources of data, our primary sample consists of 305 UAs in the United States over the period of 2006–2017 with 35,781 UA-month observations.

The contribution of this study is twofold. First, this study applies novel and reliable causal effect estimators rooted in econometrics, computer science, and statistics to a controversial empirical question. Unlike a conventional two-way fixed effects model, CEs take

observations under the treatment condition as missing data and directly estimate their counterfactuals. CEs convert the original causal identification problem into a “prediction” problem: Using the observations in untreated condition (without observing Uber entry) to predict the missing outcomes of treated observations (observing Uber entry) in counterfactual condition (Athey et al. 2021). Among the three CEs introduced, IFect and MC estimators attempt to condition on unobserved, time-varying confounders using a latent factor approach. Therefore, estimates from IFect and MC are more reliable causal effects than the conventional method when the assumption of no time-varying confounders appears to fail (Athey et al. 2021, Liu et al. 2020). Also, our study also contributes to a new but rapidly growing literature on machine learning (ML) and causal inference. This new stream of literature takes many of the strengths and innovations of computational methods to solve causal inference problems (Athey 2019). There are a variety of different problems that can be tackled with ML methods, and a panel data setting (including DID designs) is one of them. We apply a set of CEs to an actual empirical problem and provide a practical direction on the future application of CEs. To the best of our knowledge, our study is the first to apply CEs to strengthen casual identification in OM and Information Systems literature.

Our second contribution is a series of empirical evidence on Uber's effect on public bus transportation. The finding from CEs contributes to the debate on substitution or complementarity of ride-sharing platforms on traditional public transit. A number of empirical papers examine this question (Babar and Burtch 2020, Graehler et al. 2019, Hall et al. 2018), but no consensus has been reached, which could be due to the strong assumption of the absence of time-varying confounders. Our evidence from CEs lends support to Uber's substitution effect on the public bus transit systems. Next, we also add empirical evidence that Uber has not significantly affected public demand response transportation, which indicates that Uber is not directly competing with other transportation services that aim to solve the “last-mile problem.” Recall that the main supporting argument for the existence of a complementary effect on public transportation is that Uber can connect riders to transportation hubs, such as bus stops, and help solve the “last-mile problem.” If this argument is true, we should observe that Uber directly competes with the other transportation services that aim to solve the “last-mile problem.” However, our result suggests that Uber does not seem to compete with the other transportation services that aim to solve the “last-mile problem,” which further supports that a complementary effect on public transportation is unlikely to be true. Further, our study

adds to the literature by providing a deeper understanding of how Uber is coexisting with public transportation systems. In particular, the negative impact of Uber entry on public transportation systems is heterogeneous. In areas with a larger population using public bus transportation as a routine mode to commute to work, the negative impact is weaker. In areas with a large population under poverty, with disabled people, people with a language barrier, or a large unemployment population, the negative impact is amplified. Overall, all these empirical findings can be coherently explained in our framework of the substitution effect of Uber.

2. Literature Review

In this section, we review the literature from four aspects: (i) sharing economy operations; (ii) transport OM; (iii) the impact of ride-sharing platforms on public transit operations; and (iv) causal inference with ML or computational statistic methods for panel data. We highlight our contributions by comparing and contrasting our work with past studies.

2.1. Sharing Economy Operations

Prior studies on sharing economy operations have focused on three main areas: on-demand access to services in sharing economy, the effect of sharing economy on labor supply, and the effect of sharing economy on various societal and economic outcomes. The first stream has investigated how on-demand access to services in sharing economy extends the existing OM theory (Benjaafar and Hu 2020). The second stream has examined new features of labor supply in the sharing economy. Chen et al. (2017) showed that Uber drivers benefit significantly from real-time flexibility in both setting a customized work schedule and also adjusting it throughout the day. Angrist et al. (2017) compared the differences in the compensation arrangements available to traditional taxi and ride-sharing drivers and evaluated the economic value of ride-sharing work opportunities for drivers. Cook et al. (2018) found that although the sharing economy brings more flexibility in employment, there still exists a significant gender earning gap among Uber drivers. Our study examines the disruptive effect of the sharing economy on public transportation operations.

The third stream of literature has focused on the impact of the introduction of sharing-economy platforms. Cramer and Krueger (2016) showed that riding sharing platforms, such as Uber and Lyft, are providing unprecedented competition in the taxi industry: Uber drivers achieve significantly higher capacity utilization rates than taxi drivers. Greenwood and Watal (2017) estimated how the entry of Uber influences the rate of alcohol-related motor vehicle fatalities.

Zervas et al. (2017) found that the entry of Airbnb, a prominent platform for short-term accommodations, has a negative causal impact on hotel revenue. Nie et al. (2019) examined the entry of Airbnb on the review manipulation activities of hotels. Mayya et al. (2020) explored the economic benefits of forgoing screening on the Airbnb platform. Zhang et al. (2017) demonstrated the impact of image factors on property demand in Airbnb. Gong et al. (2017) examined the impact of Uber entry on expensive vehicle purchases. Burtch et al. (2018) investigated how Uber entry influences local entrepreneurial activity. A more recent paper discusses the substitution effects of bike-sharing on ride-sharing platforms regarding solving the last-mile transportation problem (Qin et al. 2018). Our study complements this stream by examining the impact of Uber on public transportation and considering the impact across various societal and economic conditions.

2.2. Transport OM

Our study is also related to the literature on transport OM. Naumov et al. (2020) examined the impact of automated vehicles on public transit systems using an analytical model. Hao et al. (2020) considered the idle vehicle pre-allocation problem with uncertain demands in the context of the taxi industry. Freund et al. (2018) proposed a data-driven method to allocate capacity in bike-sharing systems. Traffic logistics and emission have received attention in the OM literature (Qi and Shen 2019, Van Woensel et al. 2001). Jabali et al. (2012) examined CO₂ emissions in a time-dependent vehicle routing context. Our study contributes to this stream of literature by investigating the impact of the sharing economy technology on public transit operations.

2.3. Ride-Sharing Platform and Public Transit

Our study is closely related to studies on the effect of the sharing economy on public transportation. There are a few papers examining the effects of ride-sharing platforms on various transportation modes. We provide an overarching unification of these studies in Online Appendix A (see Table A1). Next, we discuss the details of each study. Aiming at understanding whether ride-sharing services affect traffic congestion, Li et al. (2021b) used urban mobility and congestion statistics for the 101 UAs in the United States from 1982 to 2014 to demonstrate that the entry of Uber significantly decreased traffic congestion time, congestion costs, and excessive fuel consumption. Further, there are a couple of papers devoted to understanding the complex relationship between public transit and ride-sharing platforms. Similar to the debate from practitioners, mixed evidence is presented in academic research. Clewlow and Mishra (2017) found that people using ride-sharing services are less likely

to use public transit. This study is based on an extensive self-administered travel and residential choice survey, drawing on questions commonly used in the regional transportation surveys and previous travel behavior research. They administered the survey using a targeted email approach and more than 4000 responses were collected. However, the sample used in the study is only limited to seven major metropolitan areas in the United States.

Next, with a transit agency-month level panel data, Hall et al. (2018) found that Uber has complemented public transit, and Uber increases ridership more in larger cities and for smaller transit agencies. Graehler et al. (2019) argued that Hall et al. (2018) were likely incorrect because they found that the introduction of bike share in a city was associated with an increase in light and heavy rail ridership and a decrease in bus ridership, but for the ride-sharing platform, referred to as transportation network companies, both heavy rail ridership and bus ridership decreased. They conducted a longitudinal analysis at the transit agency-month level, and the data are from 22 transit agencies from January 2002 to April 2018.

There are also several drawbacks in the prior literature, which leaves space for our study. First, the study of Graehler et al. (2019) was based on data at the transit agency level: 22 transit agencies are in the sample. It certainly includes core metropolitan areas with multiple transit modes, such as Boston, New York City, Houston, Dallas, Los Angeles, etc. However, it is not clear what percentage of all metropolitan areas in the United States is in the sample and how the results can be generalized.

Second, Graehler et al. (2019) adopted a random effects model as the main model, which relies on a strong assumption that the agency-specific effects are uncorrelated with the other covariates of the model. Therefore, the finding from the study is unlikely to be a causal relationship.

Third, Hall et al. (2018) and Babar and Burtch (2020) adopted a similar DID approach and verified the “parallel trend” assumption using various robustness checks. However, the main conclusions in Hall et al. (2018) and Babar and Burtch (2020) are the opposite. Hall et al. (2018) found that ride-sharing service is a complement to public transit, increasing ridership by 5% after 2 years. On the other hand, Babar and Burtch (2020) found that ride-sharing service has led to significant reductions in the utilization of city bus services. A possible reason for these conflicting results is the failure to account for the unobserved time-varying confounders sufficiently. Our study fills this gap by leveraging the recently developed CEs from econometrics, computer science, and statistics literature. The CEs rely on different functional form assumptions. IFECT CE and MC CE use a latent factor

approach to address the concern of unobserved time-varying confounders. Meanwhile, to further contribute to this complex and debatable question, our study provides comprehensive empirical evidence of the impact of the entry of Uber on public transit using observational data at each UA level. We also bring in new insights into how demographic and socioeconomic characteristics of UAs moderate the main effect of Uber entry, which provides a deeper understanding of this important research question and allows policy makers to reconsider the value of ride-sharing services.

2.4. Causal Inference with Computational Statistic Methods

The study is closely related to an emerging stream of literature on estimating average causal effects in settings with panel datasets (Athey et al. 2021, Liu et al. 2020). This stream of literature develops estimators to estimate the (average) effect of the treatment on the treated units by imputing the missing potential outcomes or counterfactuals, while the estimators differ from each other in the way of constructing treated counterfactuals.

Liu et al. (2020) discussed the properties and unified this type of estimators in a simple framework, and called them CEs. The three CEs that we use in this study are rooted in different streams of methodologies. The FEct shares the same assumptions with conventional two-way fixed effects models, while relaxing the assumption that the treatment effect is constant for all i and t . IFECT is rooted in factor-augment models: the confounders can be decomposed into time-specific factors interacted with unit-specific factor loadings (Bai 2003, 2009, Liu et al. 2020, Xu 2017). The MC estimator is established in MC literature (Athey et al. 2021, Candes and Plan 2010, Candès and Recht 2009, Mazumder et al. 2010), focusing on imputing missing elements in a large matrix. Both IFECT and MC estimators can be viewed as MC methods based on matrix factorization (Athey et al. 2021) and aim to address unobserved, time-varying confounders using a latent factor approach (Liu et al. 2020). Simulations and results from the empirical examples suggest that IFECT and MC estimators can provide more reliable causal effects than conventional methods in the presence of unobserved time-varying confounders (Liu et al. 2020). This study extends this stream of literature by applying CEs to an important but controversial empirical question and provides practice for future applications of CEs in research.

3. Data and Sample

3.1. Sample and Data Sources

The dataset we use to execute our examinations is constructed using multiple sources. To construct our

measure of public transportation, we retrieved the unlinked bus passenger trip data in the NTD from Federal Transit Administration.¹ Congress established the NTD to be the nation's primary source for information and statistics on the transit systems of the United States. Meanwhile, NTD is widely used in transportation and urban economic research (Harford 2006, Nolan et al. 2001, Palmer et al. 2004). We use the monthly ridership time-series report, which contains monthly updated service information reported from more than 300 UAs from 2002 to 2017. We adopt the Uber entry data from Li et al. (2021b), which studied the effect of Uber on traffic congestion with a similar UA level analysis (see Table 1). For each of the UAs, Li et al. (2021b) searched the official Uber newsroom as well as the major news media to identify if and when Uber entered a UA. We include a rich set of

control variables on fuel cost from the U.S. Energy Information Administration and economic status characteristics of UAs from the Bureau of Economic Analysis. Our moderating variables regarding the demographic and socioeconomic characteristics of UAs are collected from the United States Census Bureau. After assembling different sources of data, our primary sample consists of 305 UAs with Uber entry observed in 77 UAs over the period of January 2006 to December 2017. The final sample includes 35,781 UA-month observations, representing an unbalanced panel. The list of Uber entry time into different UAs is provided in Table 1.

3.2. Data

3.2.1. Dependent Variables. Our main measure of public transit is unlinked passenger trips (UPTs)

Table 1 Uber Entry Time

Urban area	Uber entry	Urban area	Uber entry
Akron, OH	Aug-2014	Madison, WI	Mar-2014
Albuquerque, NM	Apr-2014	Memphis, TN-MS-AR	Apr-2014
Allentown, PA-NJ	Jan-2015	Miami, FL	Jun-2014
Atlanta, GA	Aug-2012	Milwaukee, WI	Mar-2014
Austin, TX	Mar-2014	Minneapolis-St. Paul, MN-WI	Oct-2012
Bakersfield, CA	Jun-2014	Nashville-Davidson, TN	Dec-2013
Baltimore, MD	Feb-2013	New Haven, CT	Apr-2014
Baton Rouge, LA	Jul-2014	New Orleans, LA	Sep-2014
Birmingham, AL	Feb-2015	New York-Newark, NY-NJ-CT	May-2011
Boise City, ID	Oct-2014	Oklahoma City, OK	Oct-2013
Boston, MA-NH-RI	Sep-2012	Omaha, NE-IA	May-2014
Bridgeport-Stamford, CT-NY	Apr-2014	Orlando, FL	Jun-2014
Charleston-North Charleston, SC	Jul-2014	Pensacola, FL-AL	Dec-2014
Charlotte, NC-SC	Sep-2013	Philadelphia, PA-NJ-DE-MD	May-2012
Chicago, IL-IN	Sep-2011	Phoenix-Mesa, AZ	Oct-2012
Cincinnati, OH-KY-IN	Mar-2014	Pittsburgh, PA	Mar-2014
Cleveland, OH	Apr-2014	Portland, OR-WA	Dec-2014
Colorado Springs, CO	May-2014	Providence, RI-MA	Sep-2013
Columbia, SC	Jul-2014	Richmond, VA	Aug-2014
Columbus, OH	Dec-2013	Sacramento, CA	Jan-2013
Corpus Christi, TX	May-2014	Salt Lake City-West Valley City, UT	May-2014
Dallas-Fort Worth-Arlington, TX	Sep-2012	San Angelo, TX	Mar-2014
Dayton, OH	Aug-2014	San Francisco-Oakland, CA	Jun-2012
Denver-Aurora, CO	Sep-2012	Seattle, WA	Aug-2011
Detroit, MI	Mar-2013	Spokane, WA	May-2014
El Paso, TX-NM	Jun-2014	Springfield, MA-CT	Apr-2015
Fresno, CA	Feb-2014	St. Louis, MO-IL	Oct-2014
Grand Rapids, MI	Jul-2014	Stockton, CA	May-2014
Greensboro, NC	Jun-2014	Tampa-St. Petersburg, FL	Apr-2014
Houston, TX	Feb-2014	Toledo, OH-MI	Jun-2014
Indianapolis, IN	Jun-2013	Tucson, AZ	Jun-2014
Jackson, MS	Dec-2015	Tulsa, OK	Mar-2014
Jacksonville, FL	Jan-2014	Urban Honolulu, HI	Dec-2013
Kansas City, MO-KS	May-2014	Virginia Beach, VA	May-2014
Knoxville, TN	Aug-2014	Washington, DC-VA-MD	Nov-2011
Las Vegas-Henderson, NV	Nov-2014	Wichita, KS	Aug-2014
Little Rock, AR	Nov-2014	Winston-Salem, NC	Jun-2014
Los Angeles-Long Beach-Anaheim, CA	Mar-2012	Worcester, MA-CT	Oct-2014
Louisville/Jefferson County, KY-IN	Apr-2014		

Note: This table shows the 77 urban areas with Uber entry in the main sample.

for buses, which are counted each time passengers board public transportation vehicles (meaning buses), regardless of how many buses they use to travel from their origin to their destination. In our sample, the mean UPT is 1539.43 thousand trips each month, as presented in Table 2. The second variable to measure public transit is UPTs standardized by the vehicle revenues hours (VRH) in that month. The VRH is the hours that vehicles are scheduled to actually travel while in revenue service, which includes layover and recovery time but excludes operator training, vehicle maintenance testing, etc. We report the statistics of VRH by month in Table 2. The statistics indicate that the average VRH is 45.25 thousand hours per month. In the falsification test, we also use UPT and VRH for Demand Response (which is known as paratransit), including services where a transit vehicle does not operate a fixed route or on a fixed schedule but picks up and drops off passengers at select locations in response to specific service requests.

3.2.2. Control Variables. We control for a vector of socioeconomic factors that may affect the demand for bus transportation systems in a UA, such as average gasoline cost and average diesel cost. Additionally, we control for the local economic characteristics, including the GDP, population, and per capita income of a UA. Table 2 describes the variables as well as the

summary statistics of these variables. Table 3 presents the correlation table for main variables.

3.2.3. Moderating Variables. In addition to the direct effect of Uber entry, we also test a set of moderating variables to evaluate how Uber's effect on public transportation is varying due to the local demographic and economic status. We collect our moderating variables from 2010 to 2015, which covers every single time period when Uber entry to UAs in our sample is observed. We briefly introduce the moderating variables here and discuss the rationale and theoretical framework in detail in section 5. The first group of variables we examine is related to *commuting mode to work*. Mean travel time to work, the population needs to travel to work, various modes of travel (such as public transportation, drive alone, and carpool), as well as the percent of work from home jobs, are examined. Second, we consider the group of *disability and language ability*. We collect data on the population with a disability, with a disability under 18, and with a disability between 18 and 64. Further, we collected data on the number of people who spoke a language other than English at home and the number of people who spoke English less than "very well." The third set of moderating variables is related to *unemployment*. We obtain data on the total number of unemployed, self-employed workers in their own incorporated business, and the number of unpaid

Table 2 Definition and Summary Statistics of Main Variables

	Definition	Obs	Mean	SD	Min	Max
UPT_Bus	Unlinked passenger trips with bus (thousands)	35,781	1539.43	7125.63	0.58	126,341.8
VRH_Bus	Vehicle revenue hours with bus (thousands)	35,781	45.25	160.40	0.00	2462.14
UPT_DR	Unlinked passenger trips with flexible demand response (thousands)	36,711	26.56	67.55	0.00	916.41
VRH_DR	Vehicle revenue hours with demand response (thousands)	36,711	13.64	41.39	0.00	682.16
GDP	GDP in dollars (\$1000)	35,781	49.95	123.79	2.27	1657.46
POP	Population (thousands)	35,781	909.55	1842.53	70.81	20,153.63
PCI	Per capita income (\$1000)	35,781	39.35	8.35	18.67	106.67
Diesel cost	Average diesel cost (\$/gallon)	35,781	3.25	0.68	1.88	4.97
Gas cost	Average gasoline cost (\$/gallon)	35,781	3.01	0.61	1.62	4.53

Table 3 Correlation Table

	1	2	4	3	5	6	7	8
1. UPT_Bus	1.000							
2. VRH_Bus	0.988*	1.000						
3. UPT_DR	0.892*	0.921*	1.000					
4. VRH_DR	0.939*	0.960*	0.970*	1.000				
5. GDP	0.912*	0.948*	0.928*	0.938*	1.000			
6. POP	0.893*	0.935*	0.929*	0.927*	0.983*	1.000		
7. PCI	0.274*	0.311*	0.348*	0.336*	0.406*	0.355*	1.000	
8. Diesel cost	0.024*	0.018*	0.024*	0.019*	0.017*	0.018*	0.064*	1.000
9. Gas cost	0.028*	0.023*	0.031*	0.021*	0.022*	0.024*	0.056*	0.940*

Note: *t* statistics in parentheses, * $p < 0.05$.

Table 4 Definition and Summary Statistics of Moderating Variables

Variables	Definition	Mean	SD	Min	Max
Unemployment					
Unemployed (million)	Total number of unemployed	0.094	0.130	0.010	1.014
Self-employed workers (million)	Self-employed workers in own not incorporated business	0.057	0.083	0.008	0.575
Unpaid family workers (million)	Unpaid family workers	0.001	0.002	0.000	0.019
Disability and language ability					
With disability (million)	Total civilian noninstitutionalized population with a disability	0.227	0.257	0.050	2.031
With disability under 18 years (million)	Total civilian noninstitutionalized population with a disability and under 18	0.018	0.020	0.002	0.146
With disability from 18 to 64 (million)	Total civilian noninstitutionalized population with a disability and between 18 and 64	0.117	0.125	0.025	0.956
Other language at home (million)	Population 5 years and over who spoke a language other than English at home	0.503	1.061	0.018	7.404
Less well in English (million)	Population 5 years and over who spoke English less than “very well”	0.209	0.470	0.006	3.242
Poverty rate					
All families	Poverty rates for all families	11.043	3.246	5.000	33.100
Married-couple family	Poverty rates for a married-couple family	5.252	2.444	1.600	27.400
Female householder family	Poverty rates for a single-female householder family	29.643	5.437	14.400	51.900
Commuting mode to work					
Mean Travel Time to Work (minutes)	Average travel time to work	24.906	3.099	18.200	36.300
Population commuting to work (million)	Population of workers 16 years and over commuting to work	0.960	1.243	0.185	9.584
Commuting to work with public transportation (%)	Percent of workers 16 years and over commuting to work with public transportation	3.033	4.083	0.100	31.500
Commuting to work—drove alone (%)	Percent of workers 16 years and over commuting to work by driving alone	79.220	5.559	49.800	87.800
Commuting to work—carpooled (%)	Percent of workers 16 years and over commuting to work by carpool	9.359	1.670	5.700	15.700
Employment-weighted WAH Score	Percent of jobs can plausibly be performed at home weighed by employment*	0.283	0.050	0.176	0.474
Wage-weighted WAH Score	Percent of jobs can plausibly be performed at home weighed by wage*	0.333	0.051	0.216	0.511

Note: *We use two Work at Home (WAH) scores provided by Dingel and Neiman (2020).

family workers. The final group of moderating variables is the *poverty rate*. We collect poverty rate data based on family types: all family, married-couple family, and female householder (no husband present) family. The detailed description and summary statistics regarding all moderating variables are presented in Table 4. Online Appendix D presents the correlation matrix of all moderating variables.

4. The Effect of Uber Entry on Public Bus Transportation

4.1. Conventional Two-Way Fixed Effect Estimators

Following prior literature on Uber's entry effects (Burtch et al. 2018, Greenwood and Wattal 2017, Li et al. 2021b), we start with a conventional DID fixed effects method as our baseline model to examine the difference in demand for public transportation systems before and after Uber entry across multiple areas. The DID estimation has become an increasingly popular method to identify causal relationships. Meanwhile, we perform a Hausman test to verify our choice of the fixed effects model as our baseline model against the random effects model

(Wooldridge 2010). The result of the Hausman test is in favor of the fixed effects model (please see the detailed discussion in Online Appendix C.1). More specifically, we estimate the following panel data model for UA i and month t :

$$\log(\text{PublicTransit}_{i,t}) = \beta \times \text{UberEntry}_{i,t} + \gamma \times \text{Controls}_{i,t} + \alpha_i + \eta_t + \varepsilon_{i,t}, \quad (1)$$

where $\log(\text{PublicTransit}_{i,t})$ represents the natural log transformation of main dependent variables—unlinked passenger trips with buses (UPT_Bus) and unlinked passenger trips standardized by vehicle revenue hours (UPT_Bus/VRH_Bus)—for UA i in month t . $\text{UberEntry}_{i,t}$ is a binary variable, which equals to 1 if UA i has Uber service in month t , and 0 otherwise. The parameter β is our main interest. Control variables for the local economic characteristics, such as the GDP, population, and per capita income, are measured annually but provide enough variation for our model estimation. We also control for time-invariant unobservable UA-specific characteristics by including UA fixed effects α_i . η_t is a set of month dummies we use to control for economy-wide shocks. We estimate the models using OLS

regressions with robust standard clustered at the UA level to control for serial correlation (Wooldridge 2010). $\varepsilon_{i,t}$ represents the idiosyncratic error.

Columns (1) and (2) in Table 5 present the regression results of model (1). We use log transformation of UPT_Bus as the main dependent variable and an alternative measure of public transportation systems by standardizing UPT_Bus/VRH_Bus in columns (1) and (2), respectively. The coefficients of *UberEntry* are negative and significant at the 0.01 level in both columns. This finding suggests that Uber entry is negatively related to the number of passenger trips by buses in local UAs. The marginal effect calculations with the log of passenger trips with buses (column 1) show that, on average, Uber's entry is associated with an 8.78% drop in the number of passenger trips with buses. To illustrate the magnitude of the effect of Uber's entry on public transportation, an 8.78% decline in the total number of passenger trips translates to a reduction of 135.16 thousand bus trips based on the mean level of UPTs of buses (1539.43 thousand trips) in our sample, which equals to a loss of \$0.27 million if we assume that the average bus ticket is \$2 for each passenger trip. The economic effect is associated with a 3.31% drop in the number of passenger trips after standardizing by vehicle revenue hours, when all other things being equal. Taken together, this set of baseline results provide empirical evidence that Uber entry significantly decreases the number of trips by buses in the UA in the United States.

Table 5 Conventional Two-Way Fixed Effects Estimators of Uber Entry on Passenger Trips of Bus

Outcome	log(UPT_Bus)	log($\frac{\text{UPT_Bus}}{\text{VRH_Bus}}$)	log(UPT_Bus)	
	Full sample (1)	Full sample (2)	PSM matched sample (3)	CEM matched sample (4)
Uber entry	−0.088*** (0.005)	−0.033*** (0.004)	−0.032*** (0.007)	−0.063*** (0.007)
Diesel cost	−0.029 (0.026)	−0.019 (0.018)	−0.021 (0.029)	0.007 (0.032)
Gas cost	−0.126*** (0.017)	−0.025** (0.012)	−0.206*** (0.020)	−0.147*** (0.021)
log(GDP)	0.051 (0.048)	0.015 (0.033)	−0.178*** (0.068)	0.022 (0.055)
log(POP)	0.367*** (0.103)	−0.207*** (0.060)	0.722*** (0.152)	0.617*** (0.120)
log(PCI)	−0.473*** (0.079)	−0.363*** (0.056)	−0.944*** (0.110)	−0.668*** (0.092)
UA FE	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes
No. of UAs	305	305	128	229
Observations	35,781	35,746	16,571	27,875
R ²	0.978	0.858	0.979	0.965

Notes: Robust standard errors in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

One issue with the prior fixed effect estimations is that they might be driven by the difference in demographic and socioeconomic status between UAs. To address this potential problem, we execute matching methods to limit the *ex ante* differences between treated UAs and untreated UAs (Bapna et al. 2019, Cheng et al. 2020, Khurana et al. 2019, Kumar et al. 2018, Wang et al. 2021a). Then, based on the matched sample, we rerun our fixed effects model. To the extent that this procedure increases the homogeneity between the two groups, it also increases the strength of the causal claims from the change in the treatment, which is Uber entry. Specifically, we perform propensity score matching (PSM) by pairing nearest neighbor 1-on-1 for each UA in which Uber entry is observed (treated) and UA in which Uber entry is never observed (untreated) based on a set of different variables at the time point of 1 year before Uber entry.² The criteria include: first, the operation cost of public transportation systems, including gas cost and diesel cost; second, local economic status, including GDP, per capita income, and total population.³

In PSM, the propensity scores depend critically on the logit functional form. In order to obtain robust results, we also perform coarsened exact matching (CEM) based on similar settings again. CEM is a monotonic imbalance bounding matching method, which means that the balance between the treated and control groups is chosen by the user *ex ante* rather than discovered through the usual process of checking (Iacus et al. 2012). Consequently, after executing the matching procedure, we obtain a matched sample of treated and untreated UAs with a number of 128 for PSM and 229 for CEM, estimate Equation (1) again, and report the results for PSM and CEM in columns (3) and (4) in Table 5, respectively. Both the coefficients of *UberEntry* are consistently negative and significant, indicating a strong and significant substitution effect of *UberEntry* on public bus transportation.

4.2. Counterfactual Estimators

Next, to further address the concern of unobserved time-varying confounders, we adopt CEs, which leverage the recently developed computational method in causal effects estimation in panel data, to infer the causality. Essentially, the CEs convert the original causal identification problem to a prediction problem, and a critical step is to build the predictive model for estimation. In general, there are three steps in the estimation using CEs: (1) treat treated observations as missing values and build the predictive model with only untreated observations; (2) predict counterfactual outcomes for treated observations (missing value) from predictive models obtained in step (1); (3) obtain ATT by taking an average of the

differences between observed outcomes and counterfactual (predicted) outcomes for the treated observations. However, the CEs differ in the modeling assumption based on how treated outcomes in counterfactual conditions are estimated (or predicted).

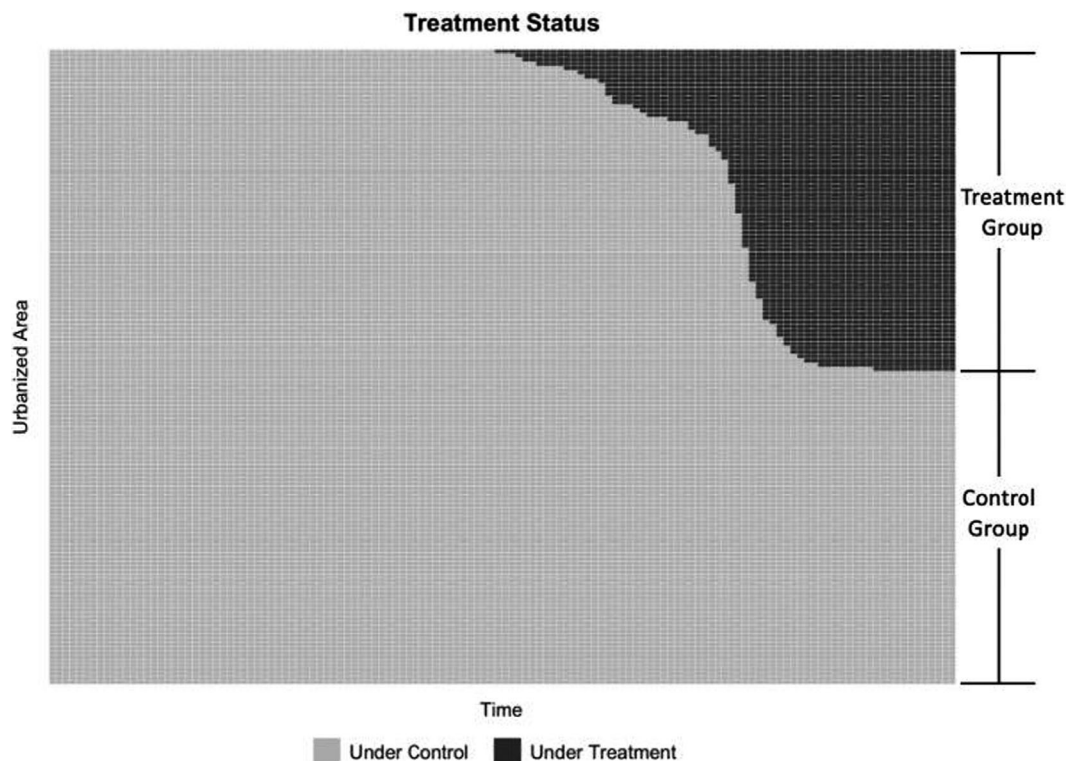
To explain the intuition, we look at our panel dataset of 305 UAs (units) and 132 months (time) period. It follows a “staggered adoption” treatment assignment mechanism. In the specific situation of our study, “staggered adoption” means Uber enters some UAs early and some UAs late. In other words, the entry time for Uber varies at different UAs. Figure 1 shows the treatment status of the first 150 UAs of our sample, with observations with Uber entry in dark gray and without in light gray. Accordingly, UAs with observed Uber entry belong to the treatment group (top panel in Figure 1), and UAs without Uber entry in any time period are assigned to the control group (bottom panel in Figure 1).

Next, the CEs proceed in the following three steps. First, CEs treat observations with Uber entry (dark cells) as missing (because we do not know the counterfactual number of bus trips if these treated UAs do not receive the treatment of Uber entry in the posttreatment stage) and use all observations without Uber (light gray cells, we know the number of bus trips when these untreated UAs do not receive the treatment of Uber entry) to build

prediction models. Essentially, we are predicting the counterfactual number of bus trips if treated UAs do not receive the treatment of Uber entry in the posttreatment stage. Depending on the different CEs, the prediction model can be a two-way fixed effects model, a factor-augmented model, or an MC model. In the second step, CEs predict the counterfactual number of bus trips if these treated UAs do not receive the treatment (using the prediction model obtained from step 1) for treated units in the posttreatment period (dark cells), and we name the predicted number of counterfactual bus trips when these treated UAs do not receive the treatment $\widehat{UPT_Bus_{it}}(0)$. Third, ATT is obtained by taking an average of the differences between observed bus trips in treated UAs, $UPT_Bus_{it}(1)$, and predicted counterfactual bus trips in treated UAs, $\widehat{UPT_Bus_{it}}(0)$, in the posttreatment stage. In short, to explain the intuition, CEs leverage all the public bus trip observations in light gray cells to predict counterfactual public bus trip numbers in dark gray cells.

Here we introduce three CEs to infer the causality between Uber’s entry and demand for public transportation: (i) FEct estimator, (ii) IFect estimator (Gobillon and Magnac 2016, Xu 2017), and (iii) MC estimator (Athey et al. 2021). The three CEs differ in building predictive models and the underlying

Figure 1 Data Structure of Uber Entry



assumptions based on how treated outcomes in counterfactual conditions are predicted. Following Liu et al. (2020), we introduce them in the order of computational complexity. In addition, we perform two diagnostic tests to gauge the validity of the identifying assumptions.

4.2.1. Fixed-Effect Counterfactual Estimator. We start with the two-way FEct estimator since it uses the procedure of conventional two-way fixed effect models and shares the same assumptions. Specifically, the estimating strategy has three steps. First, a conventional two-way fixed effect model is estimated using only non-treated observations. Second, we predict counterfactual outcomes for treated units after Uber entry (treatment) using the coefficients estimated from the first step. Finally, the ATT from the differences between actual outcomes and predicted outcomes for treated units is obtained. The conventional two-way fixed effect model is not robust when there is a significant treatment heterogeneity (De Chaisemartin and d'Haultfoeuille 2020). However, FEct is robust when the treatment effect is heterogeneous (Liu et al. 2020). We apply the FEct estimator and obtain an estimate of 0.090 with a standard error of 0.026 (Table 6 column 1), even larger than the estimate from the conventional two-way FE model (Table 5 column 1).

4.2.2. IFect Estimator. The FEct estimator relies on the assumption of no unobserved time-varying confounders as the conventional two-way fixed effect

when estimating outcome in counterfactual condition $UPT_Bus_{it}(0)$ for treated observations. An alternative way is to model unobserved time-varying confounders (latent factors) using factor-augment models when the confounders can be decomposed into time-specific factors interacted with unit-specific factor loading (Bai 2009, Liu et al. 2020, Xu 2017). Following this stream of methodology, the IFect estimator estimates a factor-augmented model (Equation 2) instead of a two-way fixed effect model using observation under the control condition to when estimating $UPT_Bus_{it}(0)$ for treated observations.

$$\log(PublicTransit_{i,t}) = \beta \times UberEntry_{i,t} + \gamma \times Controls_{i,t} + \lambda'_i f_t + \alpha_i + \eta_t + \varepsilon_{i,t}, \quad (2)$$

where f_t is a vector of unobserved common time-varying factors and λ_i is a factor of unknown factor loadings. In our context, factors can be understood as time-varying trends that affect UAs and factor loadings capture their heterogeneous impacts caused by each UA's various unobserved characteristics. Therefore, predicted $UPT_Bus_{it}(0)$ for treated observations incorporates the unobserved time-varying confounders as latent factors in the factor-augment model. The results from IFect are robust and are reported in column (2) of Table 6.

4.2.3. MC Estimator. Athey et al. (2021) developed the MC estimators from the computer science literature and used the observed elements of the

Table 6 Counterfactual Estimators of Uber Entry on Passenger Trips of Bus

Outcome: log(UPT_Bus)	Two-way fixed effect (FEct) counterfactual estimator (1)	Interactive fixed effect (IFect) counterfactual estimator (2)	Matrix completion (MC) estimator (3)
Uber entry			
Coefficient	−0.090*** (0.026)	−0.073** (0.018)	−0.080*** (0.023)
Standard error			
95% confidence interval	[−0.134, −0.041]	[−0.097, −0.032]	[−0.123, −0.038]
Diesel cost	−0.023 (0.072)	−0.044 (0.064)	−0.030 (0.074)
Gas cost	−0.124** (0.066)	−0.098* (0.055)	−0.115** (0.063)
log(GDP)	0.065 (0.281)	0.067 (0.249)	0.065 (0.244)
log(POP)	0.297 (0.667)	0.293 (0.670)	0.296 (0.646)
log(PCI)	−0.495 (0.601)	−0.494 (0.544)	−0.494 (0.528)
UA FE	Yes	Yes	Yes
Month FE	Yes	Yes	Yes
No. of UAs	305	305	305
Observations	35,781	35,781	35,781

Notes: Robust standard errors in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

matrix of control outcomes corresponding to untreated observations to predict the missing elements of the matrix, corresponding to treated observations. Different from the IFect estimator decomposing the factors into time-specific factors and unit-specific factor loadings, the MC estimator seeks to directly estimate observations in the non-treated condition as an optimization problem following matrix completion literature in computer science. Both IFect and MC use lower rank matrix approximation to make predictions for treated counterfactuals, and they do not rely on the assumption of no unobserved time-varying confounders (Liu et al. 2020). We present results from the MC estimator in column (3) of Table 6. We also apply these three CEs on the CEM matched sample in Online Appendix B.

In Table 6, we observe that the coefficients of Uber Entry are consistently negative and significant for all three CEs, lending support that the substitution effect of Uber on public bus transportation is robust. However, the magnitude of the CE estimators is slightly different from each other, with the estimates from the FEct estimator (column 1) most close to the conventional two-way fixed effect estimator. In the next section, we use diagnostics tests developed by Hartman and Hidalgo (2018) and Liu et al. (2020) to probe the validity of the absence of time-varying confounders in our sample and provide practical guidance for selecting CEs properly.

4.3. Diagnostic Tests for CEs

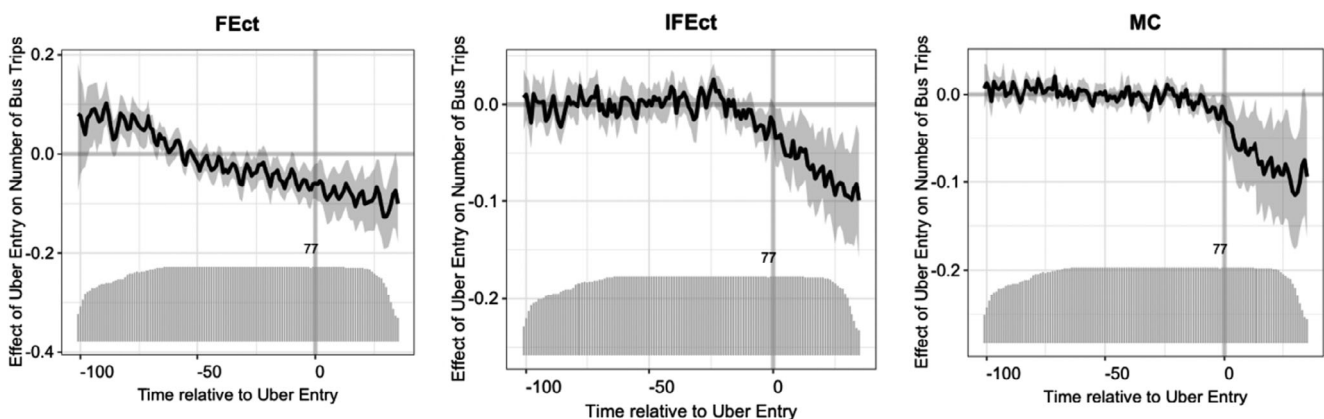
4.3.1. Dynamic Treatment Effects. Following the diagnostics procedure in Hartman and Hidalgo (2018) and Liu et al. (2020), we first plot the dynamic treatment effects based on all three CEs, then we formally run two statistical tests for the presence of

time-varying confounders. The idea of the dynamic treatment effects plot is similar to the relative time model: We examine the coefficients of the interaction terms between the treatment variable and a set of time dummy variables indicating the numbers of periods relative to the treatments. Taking advantage of the counterfactual predictions, instead of plotting the interaction terms, Liu et al. (2020) proposed plotting the averages of difference between observed outcome $UPT_{Bus_{it}}$ and $UPT_{Bus_{it}}(0)$ for units in the treatment group on the re-index time relative to Uber entry. When there are no time-varying confounders, we should expect estimates to be close around 0 in pre-treatment periods, indicating no strong “pre-trends.”

Figure 2 shows the estimated dynamic treatment effects, with 95% confidence intervals using the three CEs. From the left panel of Figure 2, we see that when using the FEct estimator, a strong downwards “pre-trend” leads toward the treatment, and multiple ATT estimates (or residual averages) in the pretreatment periods are significantly different from 0, which could be the potential reason that the conventional two-way fixed effect estimator and FEct estimator overestimate Uber’s substitution effect on public bus transportations. However, when using IFect (middle panel) and MC (right panel), we do not observe ATT estimates largely deviating from 0 in the pretreatment periods.

4.3.2. Equivalence Test and Placebo Test. Next, we formally test the presence of potential time-varying confounders with two statistical tests—*equivalence test* and *placebo test* developed by Hartman and Hidalgo (2018). Different from a commonly used Wald statistic, the null hypothesis of the equivalence test is that any ATT in the pretreatment periods falls out of a certain narrow range around 0. The rejection

Figure 2 Dynamic Treatment Effect for Counterfactual Estimators



Notes: The above figures show the dynamic treatment effects estimates using three different estimators: Fixed-effect counterfactual estimator, interactive fixed effects, and matrix completion. The histogram at the bottom of each figure illustrates the number of treated urban areas at the given time period relative to Uber entry.

of the null hypothesis implies the opposite holds with a high chance—ATTs fall in the narrow range around 0 for any pretreatment period. Intuitively, if we can show the pretreatment ATTs fall within a prespecified narrow range (the equivalence bound), we use it as a piece of evidence to support the validity of the absence of unobserved time-varying confounders. Researchers have a certain freedom to choose the prespecified equivalence bound. Liu et al. (2020) recommended setting the equivalence bound to $[-0.36\sigma_\varepsilon, 0.36\sigma_\varepsilon]$, where σ_ε is the standard deviation of residualized non-treated outcomes. Here, we choose 0.06 (close to $0.36\sigma_\varepsilon$ of IFECT estimates) consistently as equivalence bound for all equivalence tests based on the CEs.

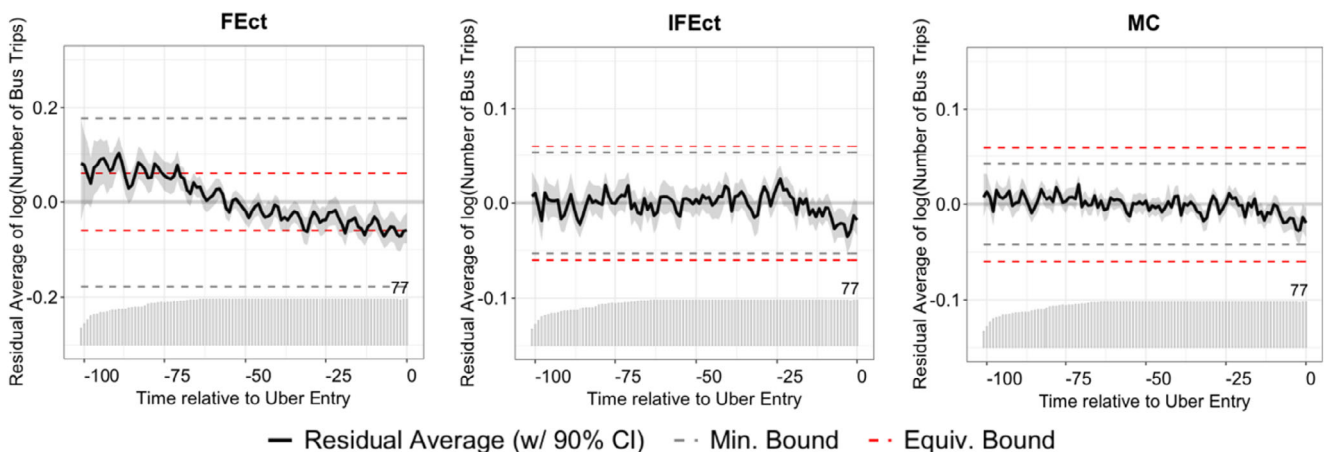
Figure 3 presents the visualized results of the equivalence test based on the three CEs: FEct, IFECT, and MC in our sample. Figure 3 also shows the equivalence bound (red dash line) and the minimal bound (gray dash line), which are determined by the largest absolute value of the range of confidence intervals of estimated ATT in the pretreatment periods. Following the rule of thumb suggested by Liu et al. (2020), when the minimal bound falls in the range of the equivalence bound, the test is considered passed. We observe that both IFECT and MC pass the test with a narrow range between minimum bounds (space between red dash lines). The confidence intervals of the pretreatment residual averages are within the equivalence bounds (gray dash lines). However, for the FEct, the confidence intervals of the pretreatment residual averages go beyond both upper and lower equivalence bounds, which results in a wide minimum bound: the minimal bound does not fall in the range of the equivalence bound. Therefore, for the FEct, the equivalence test fails to reject that the

pretreatment residual averages are not zero. In other words, unobserved time-varying confounders may bias the FEct.

Further, we conduct a placebo test, which assumes the treatment starts certain (S) periods earlier than its actual treatment event for each unit in the treatment group and apply the same CEs to obtain estimates of ATT for these S periods. The placebo test has two roles. On the one hand, it further checks if the assumption of no time-varying confounders holds. Since observations in $[-S, 0]$ periods are not exposed to the actual treatment, the estimated ATT for these S periods should be statistically insignificant from zero if the assumption of no time-varying confounders holds. On the other hand, the placebo test is designed to ensure free of the model over-fitting. Since the pretreatment data are used to build the prediction model in CEs, the over-fitting problem may exist. For example, IFECT and MC fit the non-treated data better, and the variance of the residuals is smaller, which may make the models are easier to pass the equivalence test. Note if we assume the treatment starts S periods before the actual treatment event, the data in $[-S, 0]$ periods are not used to build the predictive model. Therefore, the placebo test is immune from over-fitting because it relies on out-of-sample predictions during the placebo periods.

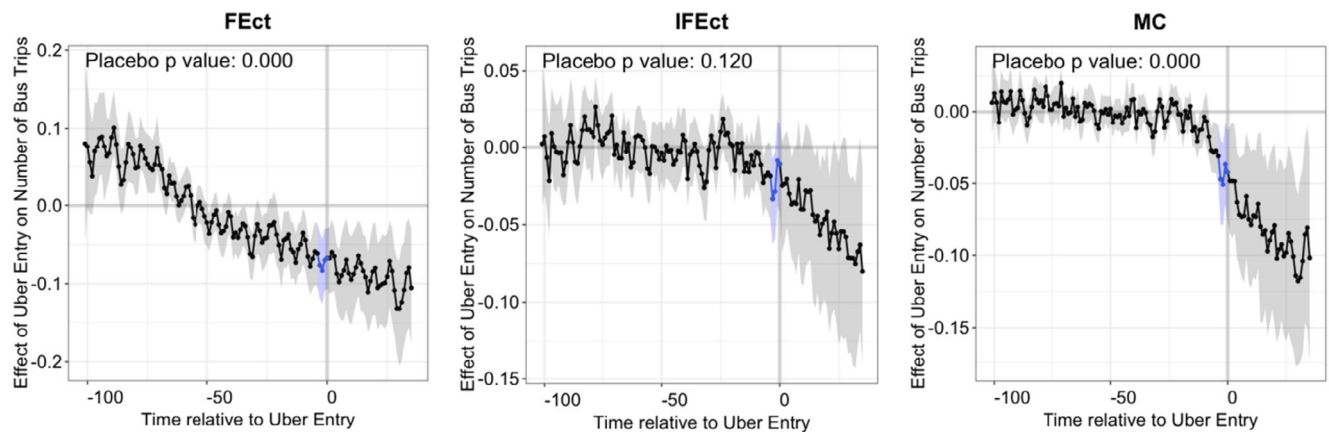
Figure 4 plots the results from the placebo tests based on the three CEs: FEct, IFECT, and MC (left, middle, and right panel) in our sample. Following Liu et al. (2020), we set $S = 3$, since the larger S is, the fewer pretreatment periods will remain to be used in estimation. We observe that IFECT passes the test ($p = 0.12$) while both FEct and MC fail the placebo test. Although the MC method passes the equivalence

Figure 3 Equivalence Tests for Counterfactual Estimators [Color figure can be viewed at wileyonlinelibrary.com]



Notes: The above figures show the results of the equivalence tests based on three different estimators: Fixed-effect counterfactual estimator, interactive fixed effects, and matrix completion. The red dashed lines mark the equivalence bound, which is set to 0.06, while the gray dashed line marks the minimum bound.

Figure 4 Placebo Tests for Counterfactual Estimators [Color figure can be viewed at wileyonlinelibrary.com]



Notes: The above figures show the results of the placebo tests based on three different estimators: Fixed-effect counterfactual estimator, interactive fixed effects, and matrix completion. Three pretreatment periods ($-3 < s < 0$) serving as the placebo are painted in blue. The p -value of the placebo test is shown at the top-left corner of each figure.

test showing the method fits the pretreatment periods well, failing the placebo test suggests that it could overfit the data due to model misspecification. Combining the results from Figures 3 and 4, IFect seems to be the best estimator, which suggests that Uber's entry is associated with a 7.3% drop in the number of passenger trips with buses.

4.3.3. Random Implementation Test. The last diagnostic test we execute is a random implementation test to rule out the possibility of the observed effect occurring purely by chance or driven by serial correlation in our dependent variable. As discussed in Bertrand et al. (2004), panel data with “staggered adoption” treatment (DID is a special case) in practice may be subject to a possible severe serial correlation problem due to the fairly long time series. In such a case, typically dependent variables are highly serially correlated, and treatment variable changes very little within a group over time. It is useful to implement the random implementation test suggested by Bertrand et al. (2004) to ensure that our estimators from CEs are not purely driven by data.

We execute this model in the following fashion. First, we randomly select approximately 25% of 305 UAs as the treated group. Next, we apply the uber entry treatment to the selected UAs using a pseudo-random number generator. However, we control that the treatment only happens after 50 time periods to ensure enough data to build the predictive estimation model to estimate the counterfactual outcomes. We then apply the three CEs on this pseudo-treatment sample and store the coefficients. We replicate this analysis for 1000 times and compare the estimate from actual treatment against the mean and standard

deviation of the pseudo-treatments. Table 7 summarizes the mean and standard deviations of the coefficient estimates for the 1000 random runs, together with the actual estimate of the Uber effect we obtain from data with the actual observed Uber entry. As can be seen, the average coefficient estimate of random assigned pseudo-treatment is relatively small and close to zero. In addition, for all CEs, the probability of a similarly sized coefficient appearing purely by chance is exceptionally low ($p < 0.001$). Finally, the real coefficient of *UberEntry* falls outside the 95% confidence interval of the pseudo-coefficient for all CEs in pseudo-treatment samples. These tests suggest that the results from CEs are not randomly driven by the computation method or the structure of our data.

4.4. Falsification Examination—Uber's Effect on Public Demand Response Transportation

Our results from both conventional estimators and CEs show a strong substitution effect of Uber entry on public transportation. Note that the main theoretical argument for the existence of a complementary effect on public transportation is that Uber can connect riders to transportation hubs and help solve the “last-mile problem.” If this argument is empirically true, and Uber indeed connects riders to transportation hubs, such as bus stops, Uber should directly compete with the current transportation services that aim to solve the “last-mile problem.” In such a case, we should observe a strong substitution of Uber on public Demand Response transportation services,⁴ which pick up passengers at selected locations in response to specific service requests and drop them off at transportation hubs, such as bus stops. For example, in Denver, FlexRide provides transportation services

Table 7 Output of Random Implementation Model with Counterfactual Estimators

Sample	Two-way fixed effect (FEct) counterfactual estimator	Interactive fixed effect (IFEct) counterfactual estimator	Matrix completion (MC) estimator
μ of random β	−0.0049212	0.00047	−0.00396
σ of random β	0.03866	0.03434	0.03642
95% confidence interval	[−0.00732, −0.00252]	[−0.00166, 0.0026]	[−0.00622, −0.0017]
Estimated real β	−0.090	−0.073	−0.080
Replications	1000	1000	1000
T-score	39.376	67.614	65.897
p-value	<0.0001	<0.0001	<0.0001

that connect passengers to other public transportation buses.

We obtain the data for Demand Response from the NTD of the Federal Transit Administration. Similar to our main dependent variable, we construct two main variables: log transformation of UPTs of Demand Response and standardized UPTs of Demand Response with vehicle revenue hours for our primary sample. Meanwhile, we construct the PSM matched sample and CEM matched sample following the same fashion when we test Uber entry's main effect on bus trips with the conventional estimator. The results from the conventional two-way fixed effect model are reported in Table 8. The coefficients of *UberEntry* are consistently insignificant for the number of Demand Response trips and positive for standardized Demand Response trips. These results suggest that the substitution effect of Uber on Demand Response does not exist. In other words, the results in Table 8 show that Uber is not directly competing with Demand Response services and cast doubts on the theoretical argument that Uber entry had a complementary effect on the public transportation system.

4.5. Considering the Effect of Lyft Entry

Finally, we consider the influence of Lyft, the second-largest ride-sharing platform, by including the Lyft dummy in the baseline model. Lyft is similar to Uber in terms of the platform function and mechanism, thus we expect a similar impact on public bus transit for Lyft. We observe a consistent substitution effect of Lyft on public bus transit, and Uber's effect is still significant after taking into consideration the influence of Lyft. More detailed results and discussions are included in Online Appendix C.2.

5. Exploring Underlying Mechanisms

5.1. Analysis Framework: Demand and Supply

The substitution effect of Uber on public bus transit systems indicates that people use a ride-sharing platform and bus transit as alternative transportation means, and a certain number of people switch from bus transit systems to a ride-sharing platform after it is available. In this section, we discuss the underlying

Table 8 Uber Entry on Passenger Trips of Demand Response Transportation

Outcome	log(UPT_DR)	log($\frac{UPT_DR}{VRH_DR}$)	log(UPT_DR)	
	Full sample (1)	Full sample (2)	PSM matched sample (3)	CEM matched sample (4)
Uber entry	−0.011 (0.018)	0.029*** (0.003)	0.016 (0.023)	−0.013 (0.023)
Diesel cost	0.072 (0.103)	−0.035** (0.015)	−0.470*** (0.152)	0.047 (0.141)
Gas cost	−0.127* (0.067)	0.005 (0.010)	−0.401*** (0.091)	−0.264*** (0.085)
log(GDP)	0.298* (0.157)	−0.245*** (0.027)	−0.139 (0.142)	0.644*** (0.182)
log(POP)	0.369 (0.321)	−0.085 (0.055)	−1.376*** (0.406)	1.971*** (0.384)
log(PCI)	−0.826*** (0.218)	−0.012 (0.039)	0.504** (0.199)	−1.133*** (0.249)
UA FE	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes
No. of UAs	311	311	128	235
Observations	36,711	35,808	16,571	28,727
R ²	0.787	0.645	0.753	0.739

Notes: Robust standard errors in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

mechanisms of the substitution effect of Uber on public bus transit systems from the overarching demand and supply aspects. To understand these mechanisms, we also further zoom in to the switching behavior for certain demographic groups.

To begin with, we give thoughts to the competitive advantages of a ride-sharing platform on the demand side. A ride-sharing platform and bus transit compete in the market of transportation. The competitive effects of sharing economy platforms have been documented in several studies, such as the impact of Airbnb on the hotel industry (Zervas et al. 2017), the impact of ride-sharing on the taxi industry (Wallsten 2015), and the labor market of ride-sharing platforms (Li et al. 2021a). If a ride-sharing platform is more competitive and attractive to certain groups, then the substitution effect is amplified in a UA with a large number of passengers belonging to this group. The reason is that people in this group are more likely to

switch to Uber after this choice is available. For example, in general, the more one needs to commute, the higher the chance one may choose Uber as a flexible commute mode than bus transit. Thus, if a UA has a large number of people who need to commute, the substitution effect of Uber would be amplified. Second, given the flexible and easy access, ride-sharing service platforms could be a more competitive choice than bus transit for people with more transit constraints, such as disability or language barriers. In this reasoning, Uber would have a increased substitution effect in UA with a large number of people with transit constraints since these people are more likely to shift to Uber.

Next, switching to Uber or the substitution effect of Uber would be accelerated if the ride-sharing service is available at a low cost to the public. We contemplate the supply-side effect from the possibility of the local population to partner with the Uber platform. If a UA has a large number of populations with high potential to partner with Uber and to become Uber drivers, it will result in a lower cost of ride-sharing services and encourage the switching behavior, and eventually accelerate the substitution effect.⁵ Take the factor of unemployment as an example. The entry of the ride-sharing platform offers an option for stable employment for the unemployed and underemployed (Burtch et al. 2018), who are more likely to partner with Uber and to become Uber drivers. If a UA has a high unemployment rate or underemployment rate, unemployed or underemployed groups could easily choose to partner with Uber, which increases the availability of Uber and lowers the cost. As a result of low cost and being widely available, the substitution effect of Uber is enlarged. In a similar vein, for those UAs with a high poverty rate, the entry of the ride-sharing platform offers an opportunity to obtain extra income, and low-income groups can be attracted to partner with the platform (Hall and Krueger 2015), which in turn increases the supply of Uber services and enlarges the substitution effect of Uber.

Next, to examine the moderating effects of the economic factors, we continue with the conventional two-way fixed effects model as in Equation (1) and add a set of interactions between *UberEntry* and the moderating variables. The model is specified as follows:

$$\begin{aligned} \log(\text{PublicTransit}_{i,t}) = & \varphi \times \text{UberEntry}_{i,t} \\ & \times \text{ModeratingVariables}_i \\ & + \beta \times \text{UberEntry}_{i,t} \\ & + \gamma \times \text{Controls}_{i,t} + \alpha_i + \eta_t + \varepsilon_{i,t}, \quad (3) \end{aligned}$$

where *ModeratingVariables_i* represents *Commuting Mode to Work*, *Disability or language ability*, *Unemployment*, and *Poverty Rate* at UA *i* 1 year before

Uber enters, α_i captures the UA-specific fixed effect, and η_t is a set of month dummies we use to control for economy-wide shocks. The parameter φ captures the moderating effect of these four groups of demographic and socioeconomic variables.

5.2. Commuting Mode to Work

Our first interest relates to the heterogeneity in the substitution effects of sharing platform entry on different commuting modes to work. People can choose various modes to commute to work. In general, the more one needs to commute, the higher the chance one may choose Uber as a flexible commute mode. We observe this pattern in the results from the moderating effects of commuting mode. In Table 9, columns (1) and (2) indicate if a UA has a longer mean travel time and a larger population needs to commute to work, the substitution effect of Uber entry is larger. Next, we zoom in three specific commuting modes: public transportation, drove alone, and carpool. When public bus transportation is used as a routine mode to commute to work, there is less demand to switch to Uber since bus transportation is much cost-efficient (column 3 in Table 9). No significant effects are observed when a larger percentage of the population drives alone to work (column 4), while the substitution effect is amplified when a larger percentage of the population uses carpool (column 5), which is expected to be the most substitutable commuting mode for Uber.

Further, we explore the feasibility of Work at Home (WAH), which requires a minimal level of commuting to work. Since regular transportation to work is not necessary for people working at home, they may share one transport vehicle with family members or do not own a vehicle at all. If they have temporary-based special needs to commute, the flexible response offered by Uber would be the best fit for such specific service requests; therefore, it enhances the substitution effect of Uber. We obtain the WAH feasibility score for each UA from Dingel and Neiman (2020), which classifies the feasibility of working at home for all occupations and merges this classification with occupational employment counts at the MSA level.⁶

Columns (6) and (7) present results from WAH scores weighted by the number of employment of occupations in a specific UA, and WAH scores weighted by the average wage of occupations in a specific UA. Negative and significant coefficients are consistent with our conjecture that the substitution effect of the sharing platform is more significant when there is a higher proportion of people working at home.

Table 9 Moderating Factor of Commuting Mode to Work

DV: Log(UPT_Bus)	Commuting to work						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Uber entry	0.023 (0.025)	−0.016** (0.007)	−0.051*** (0.007)	−0.042 (0.026)	0.140*** (0.020)	0.110*** (0.027)	0.116*** (0.033)
Uber entry * Mean travel time to work	−0.002*** (0.001)						
Uber entry * Population of commuting to work (million)		−0.015*** (0.002)					
Uber entry * Commuting to work with public transportation (%)			0.001*** (0.000)				
Uber entry * Commuting to work—drove alone (%)				−0.000 (0.000)			
Uber entry * Commuting to work—carpooled (%)					−0.020*** (0.002)		
Uber entry * Employment-weighted WAH score						−0.460*** (0.079)	
Uber entry * Wage-weighted WAH score							−0.416*** (0.082)
Diesel cost	−0.077*** (0.024)	−0.074*** (0.024)	−0.081*** (0.024)	−0.080*** (0.024)	−0.081*** (0.024)	−0.060** (0.027)	−0.059** (0.027)
Gas cost	−0.092*** (0.019)	−0.089*** (0.018)	−0.100*** (0.019)	−0.096*** (0.018)	−0.055*** (0.018)	−0.129*** (0.019)	−0.129*** (0.019)
log(GDP)	−0.325*** (0.086)	−0.308*** (0.086)	−0.344*** (0.086)	−0.336*** (0.086)	−0.430*** (0.089)	−0.365*** (0.084)	−0.365*** (0.084)
log(POP)	1.514*** (0.168)	1.491*** (0.168)	1.525*** (0.169)	1.509*** (0.168)	1.795*** (0.182)	1.864*** (0.173)	1.863*** (0.173)
log(PCI)	0.051 (0.066)	0.060 (0.066)	0.067 (0.066)	0.064 (0.066)	0.172** (0.069)	0.107 (0.070)	0.106 (0.070)
UA FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
No. of UAs	74	74	74	74	74	71	71
Observations	9737	9737	9737	9737	9737	9329	9329
R ²	0.992	0.992	0.992	0.992	0.992	0.993	0.993

Notes: Robust standard errors in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

5.3. Disability and Language Barrier

Next, ride-sharing service platforms provide more flexible and easy access to transit compared with public bus transit systems. Therefore, people with more transit constraints, such as disability or language barriers, would be more likely to shift to Uber. In addition, people with language barriers would also be more likely to turn to Uber so that there is less exposure to the public social environment as a result of traveling with public bus transit systems. To test this conjecture, we acquire data regarding the population with disabilities in different age groups as well as the population with language barriers. Table 10 reports the results. Columns (1)–(3) present the estimates with the moderating factor of the population with disabilities, population with disabilities under 18 years, and population with disabilities between 18 and 64, respectively. We observe that the coefficients of the interaction terms of *UberEntry* with these moderating factors are consistently negative and statistically significant, indicating that the ride-

sharing platform has a stronger substitution effect in UAs with a larger population with disability. We measure language barriers by the population who speak a language other than English at home and the population who speak English less than “very well.” Columns (4) and (5) report the estimates. The negative and significant coefficients of the interaction term lend further support to the idea that Uber’s substitution effect is enlarged by the population with transit constraints.

5.4. Unemployment

Third, we investigate the moderating effect of unemployment or self-employment on Uber’s substitution effect on public bus transit systems. The entry of the ride-sharing platform offers an option for stable employment for the unemployed and underemployed (Burtch et al. 2018), who are more likely to partner with Uber and to become Uber drivers. Therefore, we expect the substitution effect of Uber would be exaggerated through increasing Uber supply.

Table 10 Moderating Factor of Disability and Language Ability

DV: Log(UPT_Bus)	Disability			Language ability	
	(1)	(2)	(3)	(4)	(5)
Uber entry	−0.013* (0.007)	−0.006 (0.008)	−0.007 (0.008)	−0.026*** (0.007)	−0.027*** (0.006)
Uber entry * With disability (million)	−0.074*** (0.008)				
Uber entry * With disability under 18 years (million)		−1.131*** (0.112)			
Uber entry * With disability from 18 to 64 (million)			−0.175*** (0.018)		
Uber entry * Other Language at home (million)				−0.016*** (0.002)	
Uber entry * English less than "very well" (million)					−0.034*** (0.003)
Diesel cost	−0.075*** (0.024)	−0.074*** (0.024)	−0.074*** (0.024)	−0.075*** (0.024)	−0.075*** (0.024)
Gas cost	−0.090*** (0.018)	−0.092*** (0.018)	−0.090*** (0.018)	−0.085*** (0.019)	−0.085*** (0.019)
log(GDP)	−0.309*** (0.086)	−0.302*** (0.086)	−0.304*** (0.086)	−0.323*** (0.086)	−0.323*** (0.086)
log(POP)	1.472*** (0.169)	1.480*** (0.168)	1.473*** (0.169)	1.507*** (0.168)	1.504*** (0.168)
log(PCI)	0.062 (0.066)	0.059 (0.066)	0.061 (0.066)	0.076 (0.066)	0.076 (0.066)
UA FE	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes
No. of UAs	74	74	74	74	74
Observations	9737	9737	9737	9737	9737
R ²	0.992	0.992	0.992	0.992	0.992

Notes: Robust standard errors in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

We obtain the unemployed population, population of self-employed workers, and population for unpaid family workers from the United States Census Bureau as the measure of unemployment for local UAs. Table 11 reports the results from model (2). We note the coefficients of interaction term of *UberEntry* with unemployment or self-employment measures are consistently negative and statistically significant across all three measures in columns (1)–(3), supporting the idea that the ride-sharing platform has a stronger substitution effect in UAs with a larger population of unemployed or self-employed. Meanwhile, the economic magnitude is also meaningful. Take the estimates in column (1), for example. If the unemployed population is at a high level, say 0.16 million (0.5 SD above mean), Uber's entry will reduce approximately 3.64% of passenger trips by buses. Holding bus trips at the mean level and assuming each bus trip costs \$2, the total loss is approximately 0.113 million dollars in a month. In contrast, if the unemployed population is at a low level, for example, 28,000 (0.5 SD below mean), Uber's entry will only reduce 2.12% of passenger trips by buses, which translates to a loss of about 0.066 million dollars. In summary, we find that the substitution effect of the ride-sharing platform on

public bus transit is more pronounced for UAs with a larger population of unemployed or self-employed workers. This finding is highly consistent with previous studies showing that the Uber platform reduces lower quality entrepreneurial activity, seemingly by offering viable employment for the unemployed and underemployed (Burtch et al. 2018).

5.5. Poverty

The last moderating factor we examined is the poverty rate. The entry of the ride-sharing platform offers an opportunity to obtain extra income, and low-income groups can be attracted to partner with the platform because of the level of compensation and the flexibility it offers (Hall and Krueger 2015), which in turn increases the supply of Uber services and enlarges the substitution effect of Uber. To test this conjecture, we obtained the United States Census Bureau data of the poverty rate for different groups of families: all families, married-couple families, and single-female household families. Table 12 displays the results. We observe that our main interested coefficients for *UberEntry* * *Poverty Rate* are generally negative and significant in columns (1)–(3), showing that the high poverty rate indeed enlarges the substitution

Table 11 Moderating Factor of Unemployment

DV: Log(UPT_Bus)	Unemployment		
	(1)	(2)	(3)
Uber entry	−0.018*** (0.007)	−0.020*** (0.007)	−0.002 (0.007)
Uber entry * Unemployed (million)	−0.115*** (0.012)		
Uber entry * Self-employed workers (million)		−0.207*** (0.021)	
Uber entry * Unpaid family workers (million)			−19.458*** (1.471)
Diesel cost	−0.075*** (0.024)	−0.074*** (0.024)	−0.075*** (0.025)
Gas cost	−0.087*** (0.019)	−0.085*** (0.019)	−0.091*** (0.019)
log(GDP)	−0.313*** (0.086)	−0.313*** (0.086)	−0.309*** (0.092)
log(POP)	1.476*** (0.169)	1.505*** (0.168)	1.375*** (0.177)
log(PCI)	0.064 (0.066)	0.072 (0.066)	0.071 (0.070)
UA FE	Yes	Yes	Yes
Month FE	Yes	Yes	Yes
No. of UAs	74	74	71
Observations	9737	9737	9344
R ²	0.992	0.992	0.992

Notes: Robust standard errors in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.**Table 12** Moderating Factor of Poverty

DV: Log(UPT_Bus)	Poverty rate		
	(1)	(2)	(3)
Uber entry	0.055*** (0.010)	0.047*** (0.009)	0.001 (0.016)
Uber entry * Poverty rate of all families	−0.009*** (0.001)		
Uber entry * Poverty rate of married-couple family		−0.017*** (0.001)	
Uber entry * Poverty rate of single-female householder family			−0.002*** (0.001)
Diesel cost	−0.076*** (0.024)	−0.077*** (0.024)	−0.079*** (0.024)
Gas cost	−0.089*** (0.018)	−0.071*** (0.018)	−0.099*** (0.018)
log(GDP)	−0.434*** (0.086)	−0.451*** (0.088)	−0.356*** (0.085)
log(POP)	1.569*** (0.168)	1.703*** (0.173)	1.504*** (0.169)
log(PCI)	0.150** (0.067)	0.197*** (0.069)	0.082 (0.065)
UA FE	Yes	Yes	Yes
Month FE	Yes	Yes	Yes
No. of UAs	74	74	74
Observations	9737	9737	9737
R ²	0.992	0.992	0.992

Notes: Robust standard errors in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

effect of the Uber platform. Interestingly, when we pinpoint two different types of families under the poverty rate, the effect size of the married-couple family in column (2) is much larger than that of the female householder family (no husband present) in column (3). This is not surprising, as families with both husband and wife are more like to search for opportunities to increase family income. However, this does not necessarily apply to female householder families. Female householders are more likely to be already fully occupied to support the family. Therefore, they have little extra capacity even if there is a potential opportunity to increase family income by partnering with Uber. Such that, predictably, for a single-female householder family, the increase of the substitution effect is the least among all the family types.

6. Discussion and Conclusion

In this study, we investigate the impact of Uber entry on the public bus transit systems, and how socioeconomic factors moderate this impact. There are several notable findings from our study. First, the entry of Uber strongly and negatively affects the number of passenger trips with buses, which shows that Uber has a strong substitution effect on the public bus transit systems. On average, Uber's entry is associated with an 8.8% drop in passenger trips with buses by the conventional two-way fixed effects model. A 7.3%–8.0% drop by CEs when relaxing the assumption of no unobserved time-varying confounders.

Second, our results indicate that the substitution effect is uneven across different UAs with various demographic characteristics. As a result, a larger population using public transit as routine commuting to work mitigates the substitution effect of Uber, while a larger population using carpool as main commuting to work strengthens the substitution effect of Uber. Further, a larger population under property, with disabilities or language barriers, or a larger unemployment population also exaggerates the substitution effect of Uber. The effect size of the poverty rate among different families is worthy of attention. The high poverty rate of families with some members to potentially partner with Uber augments the substitution effect to a fair large extent. In contrast, the high poverty rate of families whose members are less likely to be Uber drivers (e.g., single-female household families) adds to Uber's substitution effect, but to a much less extent. Thus, our study calls for the attention of regulators and policymakers on minority groups (e.g., single-female householder families). Although ride-sharing platforms provide a potential way to increase income for families under the poverty level, the effect

has been limited by a family's ability to partner with the ride-sharing platform.

Our findings also provide several important practical implications. First, although the debate of “friends or foe” between public transit and ride-sharing platform is inconclusive, transportation and urban planning regulators should reconsider the threats and opportunities with the entry of ride-sharing platforms, especially when the market growth for ride-sharing platforms slows down and large ride-sharing companies seek new growth opportunities depending on better competing with public transportation. A shift from public buses to Uber could also have an impact on transportation workers' quality of life in the long run. Our work enlightens policymakers and regulators on planning public transportation infrastructure in the case that ride-sharing platforms and traditional public transportation systems coexist in the future. Second, from a social welfare perspective, our findings suggest that ride-sharing platforms not only reshape the distribution of unused or underutilized assets, such as vehicles, through digital platforms to individuals who are willing to pay for the services, but also benefit unemployed and underemployed workers by providing alternative options to increase their income with a flexible working schedule. To this end, policymakers may find it valuable to consider these effects when deciding whether to accommodate these platforms. Last but not least, although the sharing economy may increase the overall social welfare, certain minority groups might not be able to fully enjoy benefits from it. A single mother might be less able to take this offer due to an overloaded burden. We encourage regulators and social workers to provide more channels to improve the social welfare of minority groups, such as single-female householder families.

Methodology-wise, we provide valuable practice for researchers who are aiming to establish causal relationships with observational time-series cross-sectional data or panel data. With the linear two-way fixed effects model as one of the most commonly used identification strategies, a useful extension is to adopt CEs relaxing the strict identifying assumption and obtain more reliable estimates for causal inference. This study presents and compares estimates from the conventional two-way fixed effects model and results from three CEs: the FEct estimator, IFect estimator, and MC estimator. To compare their performance and probe the validity of key identifying assumptions, we demonstrate two diagnostic tests for the presence of time-varying confounders. The practice in this study provides a direction for researchers to evaluate each model and make an informed decision of which one is the most suitable for their application.

Our work also bears some limitations, thus points the way for future research in this area. First, because

of the lack of availability of the Uber transaction data, we examine only the impact of Uber entry on public bus transit systems. In the future, it would be interesting to analyze the granular data on Uber trips and their effect on public transportation. Second, a future research direction is to examine the impact of Uber entry on other forms of public transportation, such as the city metro. Finally, Uber has introduced a new product, UberPool, which allows Uber passengers to share their rides with others who are taking the same route and lower the overall cost of the ride. Conceptually, UberPool is a closer competitor to traditional public transportation. Additional analysis can explore the impact of the introduction of UberPool on public transportation.

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Notes

¹<https://www.transit.dot.gov/> (last accessed: September 15, 2020).

²The results are robust for different time points, such as 6 months or a quarter before Uber entry.

³We also extend our matching variables to include unemployment, poverty rate, median age, and disability, and the results are qualitatively not different than that reported here.

⁴Demand response transit, also known as paratransit, includes services where a transit vehicle does not operate a fixed route or on a fixed schedule except, perhaps, on a temporary basis to satisfy a special need, but picks up and drops off passengers at select locations in response to specific service requests. Demand Response transit is most commonly operated by private companies under contract with public transit agencies, but can be also operated by community groups, nonprofit organizations, or the public transit agency directly.

⁵Before we formally test moderating factors on increasing the number of potential Uber drivers, we first conduct an analysis on Uber's substitution effect over time. The number of Uber vehicles in an area could increase over time with the awareness of Uber platform. We indeed find a time trend on the substitution effect of Uber related to the increased number of Uber vehicles, given that the public transit systems remain relatively stable. The detailed results and discussion are included in Online Appendix C.3.

⁶The WAH feasibility of a certain occupation is based on responses to two Occupational Information Network (O*NET) surveys covering “work context” and “generalized work activities.” If answers to those surveys reveal that an occupation requires “daily work outdoors” or that “operating vehicles, mechanized devices, or equipment” is very important to that occupation's performance, Dingel and Neiman (2020) determine that the occupation cannot be performed from home.

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Supporting Information

Additional supporting information may be found online in the Supporting Information section at the end of the article.

Appendix A: A Summary of Related Studies on Ride-Sharing Platforms and Public Transit.

Appendix B: Counterfactual Estimators on Matched Sample.

Appendix C: Additional Analyses and Robustness Checks.

Appendix D: Correlation Matrix of Full Set of Moderating Variables.